

DOĞU ERBAŞ

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METU Computer Engineering student focused on High-Performance Computer Graphics and Game Engine Architecture, with strong foundations in C++ and low-level systems.

EDUCATION

Middle East Technical University (METU)

B.S. in Computer Engineering

Mehmet Emin Resulzade High School

High School Diploma

Ankara, Turkey

Expected Graduation: June 2027

Ankara, Turkey

EXPERIENCE

METU Robotics and AI Technologies Research Center

June 2025 – Present

Researcher

- Developed high-performance C++ pipelines for a Lunar Rover prototype using MuJoCo physics engine, focusing on real-time physics simulation and rigid body dynamics.
- Integrated OptiTrack systems to establish 3D ground truth environments for navigation and spatial data visualization.
- Implemented real-time data logging and sensor fusion scripts to analyze rover performance in simulated and physical environments.

Europower Energy and Automation Technologies Inc.

August 2025 – September 2025

Software Intern

- Developed two full-stack applications from scratch using C#, .NET MAUI, ASP.NET Core, and SQL Server.
- Engineered a Factory Equipment Tracking System to streamline internal asset assignment and monitoring using MVVM architecture.
- Built a Transformer QC Data Management System to automate test record fetching and recording across dual-database environments.
- Collaborated with cross-functional engineering teams to test and refine applications in real-world production conditions.

PROJECTS

Ray Tracer Implementation (C++)

- Implemented recursive ray casting, Linear Algebra-based transformations, and Blinn-Phong reflection models.
- Implemented camera transformations and recursive ray casting for complex scene rendering.

Forward Rendering Pipeline (C++)

- Engineered a software-based rasterization pipeline including modeling, viewing, and projection transformations.
- Implemented culling and clipping algorithms to optimize primitive rendering before screen-space mapping.

Transformer QC Data Management System (C#, .NET MAUI)

- Built a dual-database application to automate transformer test record fetching and verification.
- Streamlined test data reporting for quality engineers, contributing to the company's digitalization efforts.

Factory Equipment Tracking System (C#, .NET MAUI)

- Developed a mobile asset management tool using MVVM architecture for real-time tracking of facility equipment.
- Designed the backend API and database schema to enhance internal workflow transparency.

TECHNICAL SKILLS & LANGUAGES

Languages: C/C++, C#, Python, SQL, HTML/CSS Lua

Tools: Linux (Ubuntu), Git, OpenGL, Vim

Interests: Cyber Security, Game Development, Computer Graphics

Currently Learning: Unity, CUDA

Foreign Languages: English (Advanced, C1/C2), German (Intermediate, A2/B1)

EXTRACURRICULAR

Competitive Swimmer. — Mountaineering and Technical Climbing enthusiast. — Analog & Digital Photographer.